

Forces & Equilibrium

ANSWER KEY



Force as a vector quantity

- 1 A particle has two forces acting on it: $\mathbf{F}_1 = (3, 4)$ N and $\mathbf{F}_2 = (-1, 2)$ N. Find the resultant force and its magnitude.
Resultant $= \mathbf{F}_1 + \mathbf{F}_2 = (3 - 1, 4 + 2) = (2, 6)$ N. Magnitude $= \sqrt{2^2 + 6^2} = \sqrt{40} = 2\sqrt{10} \approx 6.32$ N.
 - 2 A particle has two forces acting on it: $\mathbf{F}_1 = (5, -2)$ N and $\mathbf{F}_2 = (1, 6)$ N. Find the resultant force and its magnitude.
Resultant $= \mathbf{F}_1 + \mathbf{F}_2 = (5 + 1, -2 + 6) = (6, 4)$ N. Magnitude $= \sqrt{6^2 + 4^2} = \sqrt{52} = 2\sqrt{13} \approx 7.21$ N.
 - 3 A particle has two forces acting on it: $\mathbf{F}_1 = (-4, 7)$ N and $\mathbf{F}_2 = (9, -3)$ N. Find the resultant force and its magnitude.
Resultant $= \mathbf{F}_1 + \mathbf{F}_2 = (-4 + 9, 7 - 3) = (5, 4)$ N. Magnitude $= \sqrt{5^2 + 4^2} = \sqrt{41} \approx 6.40$ N.
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Resolving forces into components

- 4 A force of 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.
Horizontal: $50\cos(30^\circ) = 50(0.8660) \approx 43.3$ N. Vertical: $50\sin(30^\circ) = 50(0.5) = 25$ N.
 - 5 A force of 80 N acts at an angle of 50° above the horizontal. Find its horizontal and vertical components.
Horizontal: $80\cos(50^\circ) = 80(0.6428) \approx 51.4$ N. Vertical: $80\sin(50^\circ) = 80(0.7660) \approx 61.3$ N.
 - 6 A force of 100 N acts at an angle of 60° above the horizontal. Find its horizontal and vertical components.
Horizontal: $100\cos(60^\circ) = 100(0.5) = 50$ N. Vertical: $100\sin(60^\circ) = 100(0.8660) \approx 86.6$ N.
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Equilibrium of a particle under coplanar forces

- 7 A particle is in equilibrium under three forces: $(6, 0)$ N, $(-2, 5)$ N, and $\mathbf{F}$. Find $\mathbf{F}$.
For equilibrium, the sum of all forces is zero: $\mathbf{F} = -[(6, 0) + (-2, 5)] = -(4, 5) = (-4, -5)$ N.
 - 8 A particle of weight 20 N hangs in equilibrium, supported by two strings making angles of 40° and 60° with the ceiling on either side. Set up the two equilibrium equations (horizontal and vertical) using tensions T_1 and T_2 .
Horizontal: $T_1\cos(40^\circ) = T_2\cos(60^\circ)$. Vertical: $T_1\sin(40^\circ) + T_2\sin(60^\circ) = 20$.
 - 9 A particle is in equilibrium under three forces: $(3, -4)$ N, $(5, 1)$ N, and $\mathbf{F}$. Find $\mathbf{F}$.
For equilibrium, the sum of all forces is zero: $\mathbf{F} = -[(3, -4) + (5, 1)] = -(8, -3) = (-8, 3)$ N.
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Friction and the coefficient of friction

- 10** A block of mass 5 kg rests on a rough horizontal surface with coefficient of friction $\mu = 0.3$. Find the maximum friction force before the block slides, using $F = \mu R$ and $g = 9.8 \text{ m/s}^2$.
Normal reaction: $R = mg = 5(9.8) = 49 \text{ N}$. Maximum friction: $F = \mu R = 0.3(49) = 14.7 \text{ N}$.
- 11** A block of mass 8 kg rests on a rough horizontal surface. It is on the point of sliding when a horizontal force of 23.52 N is applied. Find the coefficient of friction μ , using $g = 9.8 \text{ m/s}^2$.
Normal reaction: $R = mg = 8(9.8) = 78.4 \text{ N}$. At the point of sliding, $F = \mu R$: $23.52 = \mu(78.4)$, so $\mu = \frac{23.52}{78.4} = 0.3$.
- 12** A block of mass 10 kg rests on a rough horizontal surface with coefficient of friction $\mu = 0.4$. Find the maximum friction force before the block slides, using $F = \mu R$ and $g = 9.8 \text{ m/s}^2$.
Normal reaction: $R = mg = 10(9.8) = 98 \text{ N}$. Maximum friction: $F = \mu R = 0.4(98) = 39.2 \text{ N}$.
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Objects on an inclined plane

- 13** A block of mass 4 kg rests on a smooth (frictionless) plane inclined at 25° to the horizontal. Find the component of the weight acting along the plane, using $g = 9.8 \text{ m/s}^2$.
Component along the plane: $mg \sin(25^\circ) = 4(9.8)(0.4226) \approx 16.6 \text{ N}$.
- 14** A block of mass 6 kg rests on a smooth (frictionless) plane inclined at 35° to the horizontal. Find the normal reaction force on the block, using $g = 9.8 \text{ m/s}^2$.
Normal reaction: $R = mg \cos(35^\circ) = 6(9.8)(0.8192) \approx 48.2 \text{ N}$.
- 15** A block of mass 8 kg rests on a rough plane inclined at 30° to the horizontal. Find the component of the weight acting perpendicular to the plane, using $g = 9.8 \text{ m/s}^2$.
Component perpendicular to the plane: $mg \cos(30^\circ) = 8(9.8)(0.8660) \approx 67.9 \text{ N}$.
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Connected particles and strings over a pulley

- 16** Two masses, 6 kg and 4 kg, are connected by a light inextensible string over a smooth pulley. Find the acceleration of the system and the tension in the string, using $g = 9.8 \text{ m/s}^2$.
Using $F = ma$ for the system: $(6 - 4)g = (6 + 4)a$, so $a = \frac{2(9.8)}{10} = 1.96 \text{ m/s}^2$. Tension (from the 4 kg mass): $T - 4g = 4a$, so $T = 4(9.8) + 4(1.96) = 39.2 + 7.84 = 47.04 \text{ N}$.
- 17** Two masses, 5 kg and 3 kg, are connected by a light inextensible string over a smooth pulley. Find the acceleration of the system and the tension in the string, using $g = 9.8 \text{ m/s}^2$.
Using $F = ma$ for the system: $(5 - 3)g = (5 + 3)a$, so $a = \frac{2(9.8)}{8} = 2.45 \text{ m/s}^2$. Tension (from the 3 kg mass): $T - 3g = 3a$, so $T = 3(9.8) + 3(2.45) = 29.4 + 7.35 = 36.75 \text{ N}$.
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Newton's third law and normal reaction

- 18** A box of mass 8 kg rests on a horizontal table. A person pushes down on the box with a vertical force of 15 N. Find the normal reaction force from the table, using $g = 9.8 \text{ m/s}^2$.

For vertical equilibrium: $R = mg + 15 = 8(9.8) + 15 = 78.4 + 15 = 93.4 \text{ N}$.

- 19** A box of mass 12 kg rests on a horizontal table. A person pulls upward on the box with a vertical force of 20 N (not enough to lift it). Find the normal reaction force from the table, using $g = 9.8 \text{ m/s}^2$.

For vertical equilibrium: $R = mg - 20 = 12(9.8) - 20 = 117.6 - 20 = 97.6 \text{ N}$.

- 20** A box of mass 15 kg rests on a horizontal table. A person pushes straight down on the box with an additional vertical force of 30 N. Find the normal reaction force from the table, using $g = 9.8 \text{ m/s}^2$.

For vertical equilibrium: $R = mg + 30 = 15(9.8) + 30 = 147 + 30 = 177 \text{ N}$.

A well-designed word problem: a crate on a ramp with friction

- 21** This question has two parts. A crate of mass 10 kg sits on a ramp inclined at 20° to the horizontal, with coefficient of friction $\mu = 0.25$ between the crate and the ramp. Use $g = 9.8 \text{ m/s}^2$. (a) Find the normal reaction force on the crate. (b) Determine whether the crate remains stationary, by comparing the component of gravity along the ramp with the maximum available friction force.

(a) $R = mg \cos(20^\circ) = 10(9.8)(0.9397) \approx 92.1 \text{ N}$. (b) Component of gravity along the ramp: $mg \sin(20^\circ) = 10(9.8)(0.3420) \approx 33.5 \text{ N}$. Maximum friction: $F_{\max} = \mu R = 0.25(92.1) \approx 23.0 \text{ N}$. Since $33.5 \text{ N} > 23.0 \text{ N}$, the gravity component exceeds the maximum available friction, so the crate slides down the ramp.